

Pratyansha Rana

Software Engineer | Bhopal, India

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EDUCATION

Lakshmi Narain College of Technology & Science

B-Tech CSE (AIML), 4th Semester

Bhopal, India

Expected 2028

EXPERIENCE

Cisco Virtual Internship (AI Track)

Virtual Intern

Virtual

2026

- Completed specialized training modules focused on Artificial Intelligence and Machine Learning applications within enterprise network architectures.
- Explored AI-driven automation tools, predictive modeling concepts, and intelligent data pipeline implementations.

ACHIEVEMENTS

1st Runner-Up | *Innomaker Project Showcase SGSU Bhopal*

March 2026

- Achieved 2nd place for "QUBES", received 10,000 prize for project excellence.

PROJECTS

QUBES | *React Native, Vercel, Firebase, BB84 Protocol, AES-256*

- Designed a real-time chat architecture with Firebase Firestore listeners, enabling low-latency one-to-one message synchronization across clients.
- Built a BB84-inspired quantum key exchange simulation where Alice/Bob roles, random bases, photon payloads, and sifted key indexes are synchronized through Firestore.
- Converted sifted quantum bits into a deterministic 256-bit AES key by padding/truncating bitstreams and encoding them into hexadecimal format.

LokAwaz | *TypeScript, React Native, Firebase, SQLite, YOLO*

- Implemented real-time pothole detection via a custom YOLO model within a React Native dashcam application, optimizing background execution to 5 FPS to guarantee zero UI latency and maintain thermal efficiency.
- Architected an offline-first SQLite-to-Firebase sync pipeline to capture imagery and GPS coordinates in network-deficient zones, triggering an automated LLM-verified worker dispatch protocol upon connectivity restoration.

Phantom | *Hybrid Post-Quantum Cryptography (Crystals Kyber)*

- Prototyped per-recipient encrypted messaging using client-side key generation, local private-key storage, encrypted Firestore payloads, and metadata-only audit logs to explore privacy-preserving command visibility.
- Implemented an HQ dashboard for live operational oversight, including activity feeds, chat metadata auditing, distress alert resolution, profile inspection, and chat erasure workflows without intentionally exposing plaintext message content.

Modified BB84 for Topological Quantum Systems | *Python, QKD, Topological Qubits*

- Proposed a BB84-inspired QKD modification using braiding-based basis selection for topological qubits.
- Built a Python simulation to track basis matching, fusion errors, final key length, and error rate.
- Compared simulated topological BB84 behavior against standard BB84 error-rate assumptions.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript

Frameworks & Databases: React Native, Express.js, Firebase, Supabase, MongoDB, SQLite

Developer Tools: Google Cloud Platform (GCP), Git, Vercel

Core Competencies: Data Structures & Algorithms, Distributed Synchronization, AI/ML Integration, On-Device AI (YOLO/TFLite), Quantum Cryptography (BB84, Post-Quantum)